

Lecture 02:

Systems Thinking I

Systems as Interacting Elements, Processes, and Stakeholders

Course: *Introduction to Engineering*

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Disclaimer: this lecture references material adapted from Kosky, Balmer, Keat & Wise, Exploring Engineering (5th ed.), and from the instructor's own course slides, with modifications. No authorship of third-party material is claimed.

Last Week, in One Breath

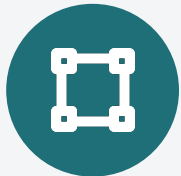
- **Engineering** = a creative, ingenious response to practical problems — not repair work.
- **It has a history:** from ancient walls and aqueducts to steam, electricity, flight, and software.
- **It has a social role:** engineers sit at the interface between science and society.
- **It has an economic role:** one of the largest, best-compensated professions worldwide.
- **It has a shared culture:** professional societies, common traits, and a code of ethics.

Agenda



What Is a System?

Elements, processes, and outputs



System Boundaries

Why the boundary is a choice



Operating Context

What constrains a system from outside



Stakeholders

Who uses, runs, and is affected by it



Your Project

Previewing Milestone 1



PART 1

What Is a System?

Interacting elements and processes, not just a pile of parts.

Defining a System

A **system** is a set of interacting elements and processes that transform inputs into outputs, existing within an operating context.

- **Elements** — The parts and people that make it up.
- **Processes** — How those parts interact and transform things over time.
- **Inputs & outputs** — What flows in, and what flows out.
- **Operating context** — The conditions the system exists within.



NOT THE SAME AS

A random pile of parts.

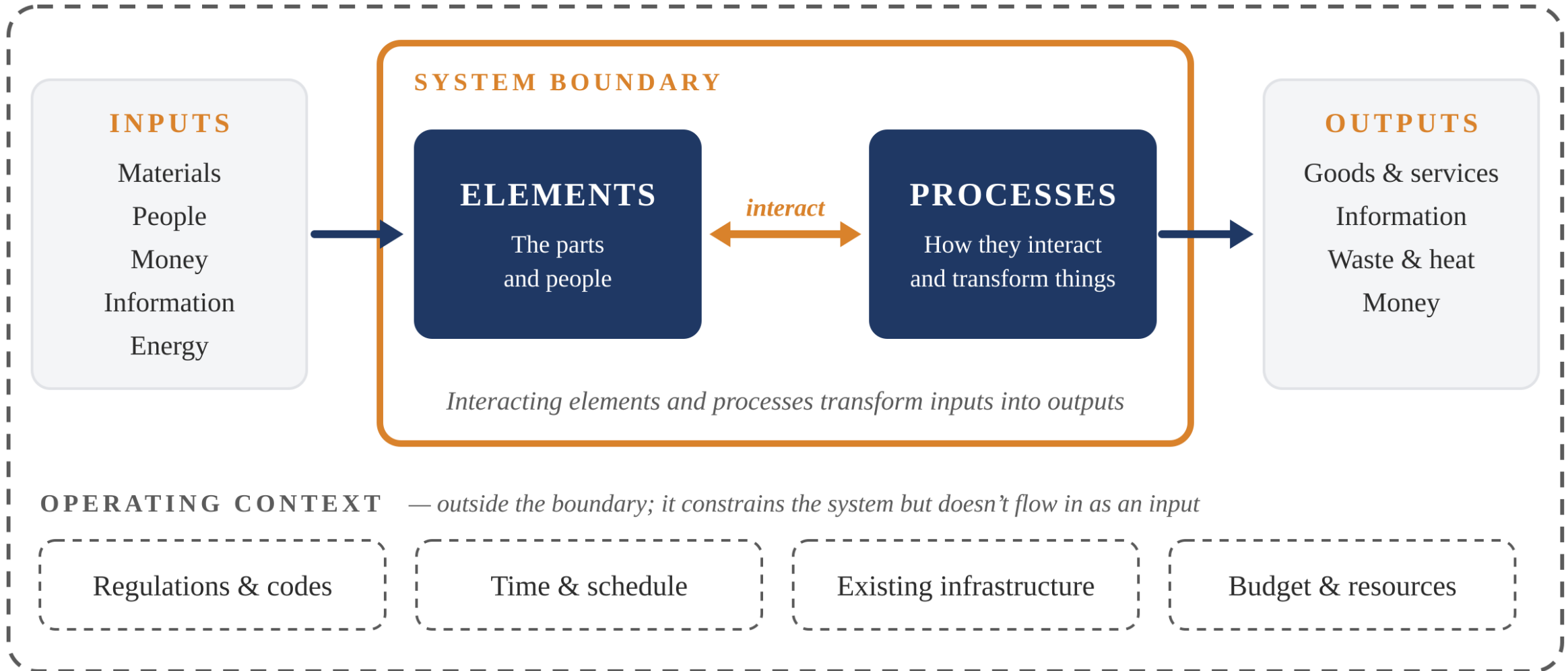


A SYSTEM

Parts with a purpose — connected so they act together, not just sit together.



The System as a Flowchart



System, or Just a Collection?

NOT A SYSTEM

A pile of bricks in a yard.

A SYSTEM

A brick wall: bricks + mortar + a load path, working together to hold weight.

NOT A SYSTEM

A student, a schedule, and a bus stop — considered separately.

A SYSTEM

The campus bus route: vehicles, schedule, and riders interacting to move people.

NOT A SYSTEM

A stove, a fridge, and a sink, sitting in a room.

A SYSTEM

A dining hall kitchen: staff, equipment, and workflow turning ingredients into meals.

NOT A SYSTEM

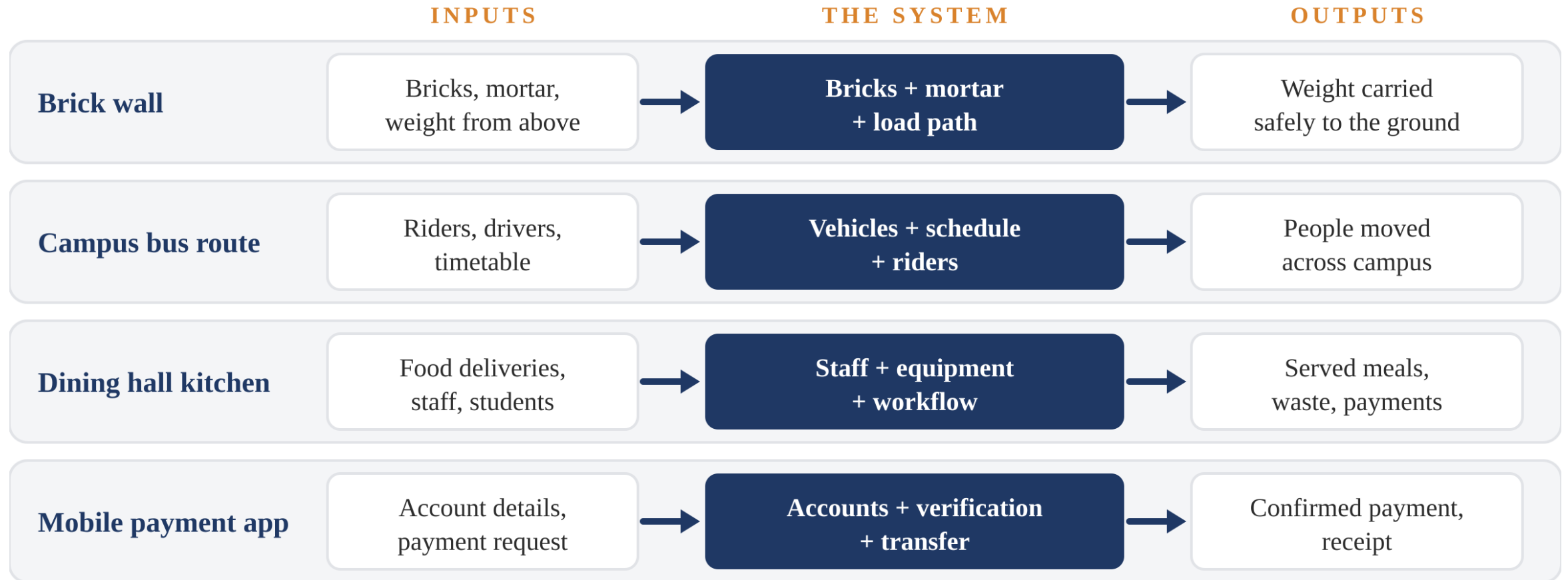
A phone, a bank, and a shop, unconnected.

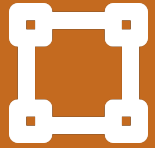
A SYSTEM

A mobile payment app: linking accounts, verification, and transfer into one flow.

WHAT IS A SYSTEM?

The Same Flow in Real Systems





PART 2

Boundaries & Operating Context

Where a system ends, and what still shapes it from outside.

The System Boundary

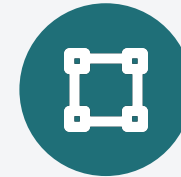
Boundary

A drawn line — often imaginary — that separates “inside the system” from “outside it.”

Anything can cross a boundary: materials, people, money, information, energy.

The key idea:

A boundary is a choice the engineer makes — not a fact about the world. Draw it tightly around one machine, or loosely around an entire building; each choice changes what you must analyze.



(Homework) Try it yourself:

Pick one of your three homework systems. Where would you redraw its boundary — tighter, or looser?

Operating Context

Just outside the boundary sits the operating context — conditions that don't cross in as inputs, but still constrain how the system can behave.



Regulations & codes

Health codes, safety standards, building codes.



Time & schedule

Opening hours, semester calendar, deadlines.



Existing infrastructure

What's already built that the system must work within.



Budget & resources

What funding, staff, or materials are actually available.

Worked Example: The Dining Hall System

INPUTS

Food deliveries →

Staff arriving →

Students entering →



System boundary

Dining Hall



OUTPUTS

→ Served meals

→ Food waste

→ Payments

Operating context: *health code · opening hours · kitchen budget · building layout*

Inside or Outside the Boundary?

- Using the dining hall example: is the supplier's delivery truck inside or outside?
- What about the university's meal-plan billing office?

50%



PART 3

Stakeholders

Everyone who uses, runs, or is touched by the system — even from outside it.

Who Are the Stakeholders?

A stakeholder is anyone who uses the system, runs it, or is affected by it — whether or not they sit inside the boundary.



Uses it

Students eating in the dining hall;
riders on the bus.



Runs it

Kitchen staff, dispatchers,
maintenance crews.



Affected by it

Neighbors dealing with noise,
delivery trucks, or traffic — often
outside the boundary entirely.

Mapping Your Stakeholders

A stakeholder map sorts every stakeholder into a simple table — a first draft is enough to start.

Uses It	Runs It	Affected By It
• Students & staff eating • Meal-plan holders • Guests / visitors	• Kitchen & serving staff • Facilities & maintenance • University dining administration	• Neighboring buildings (noise, deliveries) • Local food suppliers • Waste-management contractor

Discuss: Stakeholder Trade-offs

Stakeholders don't always want the same thing. Good systems thinking means noticing the tension, not hiding it.



Discuss

The university wants the campus bus route faster. The maintenance crew wants it safer. Riders want it cheaper.

Who are the stakeholders here — and where's the tension?

Bridge to Your Project: Milestone 1 Preview

1

Finalize your system

Pick the one system you'll study. It can be the same system that you picked at the end of Lecture 1.

2

Sketch the boundary

Draw the boundary; list at least 3 inputs and 3 outputs crossing it.

3

Map your stakeholders

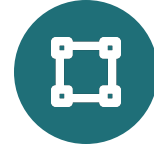
Sort them into Uses It / Runs It / Affected By It.

Due before Lecture 3: This becomes your Milestone 1 draft. Next class (Systems Thinking II) we'll turn it into a process flow diagram.

Key Takeaways



A system is elements + processes, working together with purpose — not just a pile of parts.



The system boundary is a choice the engineer makes, not a fact about the world.



Operating context shapes a system from just outside the boundary.



Stakeholders use it, run it, or are affected by it — sometimes from outside the boundary.



A stakeholder map is a simple, useful first draft — not a final answer.



Next week: turning this into a process flow diagram.

Independent Study —Milestone 1



Due before Lecture 3

Finalize your one system, and sketch its boundary and stakeholder map (Milestone 1 draft).

Examples to spark ideas: your dormitory's laundry room, the university bus route, a campus dining hall, a mobile payment app.

Why: this feeds directly into next week's process flow diagram — the more concrete your boundary and stakeholders are now, the easier that step becomes.

COMING UP NEXT

Lecture 3: Systems Thinking II

Process flow diagrams, and how different perspectives lead to different interpretations of the same system.

Thank You — Questions?

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